

Theoretical and Applied Aspects of Bank Credit Risks Minimization

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Abstract—The article deals with the theoretical and applied bases of bank credit risks management. The credit risk minimization features are defined. Methodological support of the credit risk prevention system in JSC "Kredobank" is analyzed. It has been established that bank credit risk management efficiency depends on the combination of three factors: methodological bases of financial operations analysis, bank clients and risk minimization methods. The main practical result: methodological basis recommendations on the bank's credit risks assessment formation can be used as a tool for their minimization in the personnel-oriented risk management system.

Keywords — *credit risks, risk management, analysis methods, assessment methods, methods of minimization.*

I. INTRODUCTION

Bank risk management becomes an integral element of the organization of bank's asset operations. Management of credit risks to minimize them begins with classification of loans and analysis of the bank's credit portfolio quality. Classification of loans provides for their division by certain criteria into certain categories, which differ in terms of credit risk level and amount of possible losses. The criteria for analysis and evaluation of credits, classification signs of identification of risks, evaluation of borrowers for each group of credits are not standardized, that is why each bank determines them taking into account the requirements of the NBU and international recommendations, specific economic conditions of bank functioning.

The listed features of credit risk management of the bank determine relevance of research of theoretical, methodological and practical problems in field of evaluation and minimization of risks. The research of such scientists is devoted to the problems of analysis and evaluation of risks: V. Bobyl. [1], Burdenyuk I. [2], Vasyurenko A. [3], S. Didenko. [4], Dzyublyuk A. [5], McKinsey [11], Fanelli, V. [12] and others. At the same time, there are many unresolved problems on complex use of methods of credit risk assessment taking into account norms of bank risks regulation and international practice. The purpose of the article is to describe the theoretical aspects of minimizing credit risks of the bank on the basis of applied data of JSC "Kredobank".

Risk management of the bank is the process of making managerial decisions that minimize the adverse impact of external and internal factors, as well as reduce losses caused by random events. Moreover, minimization of possible adverse impact is a priority task, solution of which does not allow a situation when the bank faces financial losses and other losses.

II. IDENTIFICATION OF BANK CREDIT RISKS

The aggregate risk of the loan portfolio depends on the riskiness of the credit transactions included in the following categories of its composition. The increase in the total risk of the bank's loan portfolio is caused by excessive concentration of (Concentration of loans in one sector of the economy), imperfect portfolio structure if its formed taking into account the needs of only customers, not the bank itself, excessive diversification, insufficient qualification of bank personnel, etc. In order to assess the aggregate risk of the loan portfolio, it is necessary to first of all to analyze the risk of all its components.

Risk identification is a preliminary stage of banking activities risk assessment, including the identification process, listing and description of such risk elements as risk sources, situations, risk factors and possible consequences. At the stage "Receipt and processing of information on risks identification" [11], the choice of risk factors is made, impact of which shall be reduced on bank activities (See Table 1).

Depending on the level of analysis, an aggregate and individual type of credit risk is identified. The aggregate risk is the risk of the credit portfolio of the bank, which provides for assessment of the volume of granted credits from the point of view of qualitative indicators of the whole portfolio. The analysis itself is performed based on the calculation of indicators that characterize the amount of outstanding payments for various categories of loans. Individual risk is the risk of a borrower that shows the amount of risk on a loan to an individual borrower [1,5]. Analysis of individual risk requires the creation of multi-variant models for its calculation, which will take into account the dependence on such factors as political, social, commercial and others.

TABLE I. THE BANK CREDIT RISKS CLASSIFICATION

<i>Criteria</i>	<i>Risks Type</i>
Level of analysis implementation	Cumulative, Individual
By area of origin	Borrower risk, credit product risk, risk of changes in the external environment of the bank and the borrower
By type of borrower	Country risk, legal entity risk, individual risk
By the nature of the manifestation	Collateral risk, moral, financial type of credit risk, business
By credit activity of the bank	Structural and procedural, personal, technological, illegal manipulation of loans, availability of credit, early payment
By type of operation	Loan, leasing and factoring operations, provision of bank guarantees and sureties, conclusion of agreements with the use of promissory notes
By degree of risk	Low, medium and high
By the degree of controllability	Localized risks (identified and controlled) non-localized risks

a. Source [4]

Depending on the sphere of occurrence, the borrower's risk, which arises in the sphere of activity of the bank's client, the credit product risk associated with the bank's functioning and the risk of changes in the external environment of the bank and the borrower are also identified. According to the type of borrower, credit risks are divided into several types, such as the risk that arises in case of foreign lending, the risk that arises in case of lending to legal entities, that arises in cases of financing activities of enterprises, banks, organizations inside the country, the risk that arises in case of lending to individuals, that arises in case of performance by a financial institution of credit operations with clients inside the country [2,3].

Risks by nature of manifestation are also divided into security risk, moral and financial type of credit risk, business risk. They belong to the circle of activity of a borrower. Moral inherent in customers who have an overvalued business reputation. Business is assessed as a result of data on development of the area where a private enterprise operates and sells its products. Financial risk arises when the profitability, composition and structure of the enterprise's assets, liquidity and the level of stability of private income are analyzed. Also, besides the above mentioned, it is necessary to highlight the risks inherent in the general credit activity of the bank: structural and procedural, personal, technological, illegal manipulations with loans, credit availability, early payment [6].

By type of operation, credit risk is divided into a risk arising in course of such operations: the bank: loan, leasing and factoring operations, provision of bank guarantees and sureties, conclusion of transactions using bills of exchange. By the nature of the borrower's actions, risks are divided into the risk of the borrower's refusal to pay interest and principal, risk of improper use of credit, and risk of impeding banking control. Credit risk indicators of Kredobank JSC in 2019/20 years are presented in Table 2.

III. METHODS OF CREDIT RISK ANALYSIS

Quantitative and qualitative are the two main types of risk analysis. On the basis of the obtained information, a methods

combination of qualitative and quantitative risk analysis is carried out. A specific feature of the qualitative approach in risk analysis is that risks are first identified, followed by a cost estimate of the consequences of the risk and development of mitigation measures. Qualitative analysis is performed at the stage of decision-making regarding financial operations of the bank. The main methods of qualitative and quantitative analysis are given in Table 2, 3.

TABLE II. CREDIT RISK INDICATORS OF KREDOBANK JSC IN 2019-2020 IN MLN UAH

<i>Credit risk indicators</i>	<i>1.03.2019</i>		<i>1.03.2020</i>	
The amount of credit debt	4049,6	993,4	5338,6	1126,9
Credit risk	243,6	36,1	356,1	45,1

a. Source [9]

TABLE III. METHODS THAT COMBINE QUANTITATIVE AND QUALITATIVE ANALYSIS

<i>Method</i>	<i>Main idea</i>	<i>Features</i>
Method of expert assessments	Expert assessment using logical and mathematical - statistical methods	The advantage of this method is the use of risk assessment software products according to standard risk sharing schemes
Creating a risk profile or risk picture	Provides an assessment of the risks of a financial transaction or a bank client on a number of parameters reflected in the group of relevant scales	It is a tool for visualizing the structure of risks and assessing the compliance of the draft decision with the bank's credit policy

b. Source [4,7].

TABLE IV. QUALITATIVE ANALYSIS METHODS

<i>Method</i>	<i>Main idea</i>	<i>Features</i>
Method of analogies and typical schemes	Provides a comparison of a number of features with the criteria and typical schemes of VK / FT	Requires completeness of information and appropriate qualifications of staff to understand the degree of its application in a particular situation
Casual analysis	Provides heuristic selection of risk events, formal and logical analysis of their possible causes and the development of minimization measures	Used at the stage of decision-making on the financial transaction for the bank's customers, stimulates the search for measures to minimize risks and increase the effectiveness of decisions
Method "Events-Conclusions"	Provides for credit risks division of into stages of decision making on crediting and credit repayment monitoring and risks identification and minimization ways for each of them	Used to identify specific risks

c. Source [6,8].

Quantitative analysis, based on probability theory tools and mathematical statistics, consists in numerical measurement of influence of changes in risk factors of credit decisions and qualitative analysis [4]. The analysis of influence of changes in risk factors of credit decisions on changes in efficiency of financial operations is based on the main variants of previous credit decisions. For the

identification of all possible risks is responsible for a qualitative analysis that determines the risk factors, the sequence of work in the performance of which arises risk and risk minimization measures. A quantitative analysis is responsible for detecting the amount of damage from various types of risk, revealing the causes, sources of risk and the amount of possible consequences. Statistical analysis and risk assessment studies the information array of losses and profits, while studying the size and frequency of the risk case [8]. On the basis of the revealed regularity, the forecast of the perspective state of profits and losses of the bank by credit operations is determined. Expert evaluations, summarizing opinions of specialists, allow determining the acceptable risk level or amount of losses.

IV. ALGORITHM OF CREDIT RISKS MINIMIZATION

Modern researchers offer various algorithms of risk assessment. Having summarized their experience, we offer (as one of the options) the following algorithm (See Fig. 1).

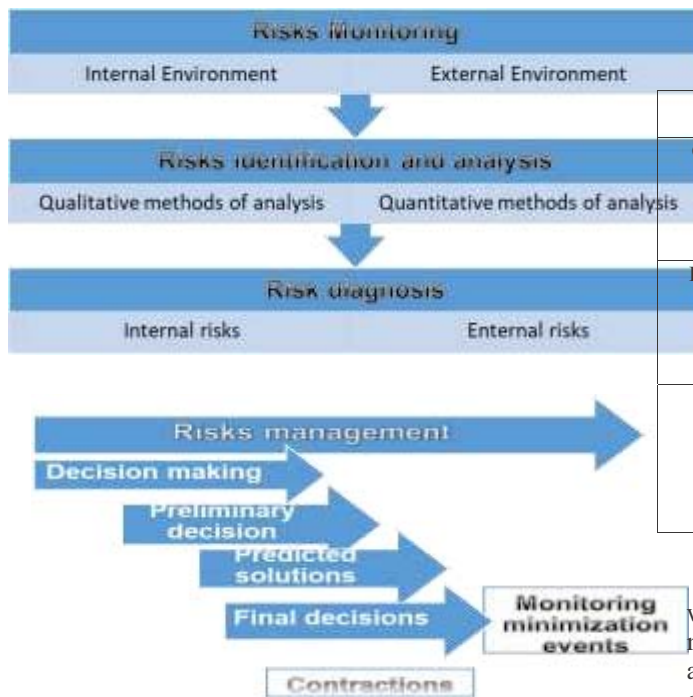


Fig. 1. A bank's credit risk minimization algorithm

Monitoring of credit risks and determination of minimization measures are performed through analysis of monthly reports on analysis of quality of the bank's credit portfolio with detailed analysis of credit risk level by the credit portfolio as a whole and by credit directions. Figure 2 shows the processes and procedures used in risk analysis.

The credit risk assessment under uncertainty conditions is reduced to determining the loan quality and the probability of its non-repayment by the borrower and analyzing the bank sensitivity to the borrower risk. These techniques can be used as tools in estimating the possible losses amount and establishing appropriate loan loss provisions in this regard. Provisioning is one of the tools to reduce bank financial risks.

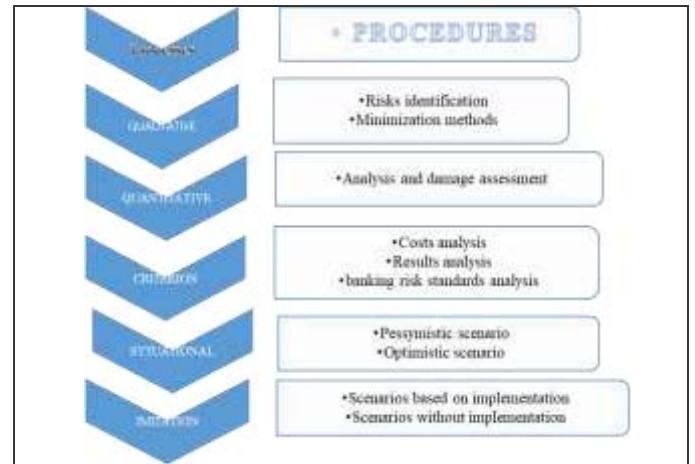


Fig. 2. Development of a bank credit risk management algorithm

To overcome the uncertainty of the credit activity of the studied bank using sensitivity analysis, which involves the use of options described in Table 4.

TABLE V. SENSITIVITY ANALYSIS OF THE BORROWER'S CREDITWORTHINESS QUALITATIVE ANALYSIS METHODS

Sensitivity options	Situation Analysis	Risks evaluations
Optimistic variant	Existing market conditions contribute to the prosperity of the borrower's business and a commercial bank can avoid losses	The probability risk realization degree is small
Pessimistic variant	When the situation in the client market and in the economy as a whole is unfavorable for the borrower's business	High risk of default by the borrower to the bank and incurring losses (ie the loan may be non-repayable)
Realistic variant	The bank accepts risks in accordance with the real situation, knowledge of the client's financial condition and credit policy of the bank	The average value of the optimistic and pessimistic options for risk assessment

d. Source [6,8].

Thus, risk management is the process by means of which the bank shows risks, assesses their amount, monitors and controls its risk positions and takes into account interrelationships between different risks categories [6,7].

V- METHODS OF BANK CREDIT RISKS MINIMIZING

There are a large number of methods to minimize risk. The main methods of minimizing the bank's credit risks are as follows: risk avoidance; distribution of risk among participants; diversification; limitation; monitoring of loan servicing; creation of flexible credit policy; creation of reserve funds; monitoring of information on identification of clients and financial operations; education and training of personnel; application of flexible technologies of risk assessment [7,8]. The main directions here are risk differentiation and loss limitation. Limitation, as an instrument of risk reduction, is intended for leveling the peaks of the bank's activity and allows setting economically justified limits of financial operations. Risk distribution (possible harm) is made between participants of economic activity in such a way that each participant is responsible for "his" risk, and possible losses of each

participant, thus, become not critical. Reservation of monetary funds or their mobilization to cover possible losses that are intended for formation of various kinds reserves do not allow distribution of responsibility for managerial decisions with other financial operations participants, but allow ensuring rhythm of bank functioning in case of the negative influence external environment increase. Each of the above-mentioned risk reduction methods also has disadvantages (See Table 5).

TABLE VI. MINIMIZING BANK CREDIT RISKS METHODS DISADVANTAGES

<i>Risk Reduction Methods</i>	<i>Consequences of Using Methods</i>
Risk allocation	Possible reduction in income
Risk insurance	Costs, limited application
Creating reserves	Low loss coverage
Hedging	Only securities risks
Diversification	Decrease in income
Limiting	Decrease in income
Avoidance	Loss of income

^c. Source [4,7]

Risk management is modeling of the deviation possibility, i.e. mismatch of planned and actually received the decision result that is caused by certain restrictions action of motivational, administrative, professionally competent, socially-psychological and physiological character. Consideration of "risk" and "risk management" categories in these positions allows transforming the approach to probability and risk consequences assessment from financially oriented to personnel oriented. The basic element of the modern risk management system is the person who needs a financial transaction and the person who makes the decision to provide a financial service. Consequently, the banking activity reducing credit risk problem is largely solved by the formation of the relevant credit policy and the use of a bank's credit portfolio assessing risks adequate methods. Such methodology can be unified partially on the basis of the credit risks bank regulation requirements. At the same time, each bank should take into account specifics of its own clientele, its own market segment, borrowers sectoral specifics, their specific capabilities.

VI. CONCLUSION

Taking into account the current understanding of the credit risk management process of the bank as part of strategic management, the following conclusions can be drawn: 1) the risk management system should correspond to the system of bank's risk assessment by regulating authorities concerning the norms of bank's risks, organization of financial monitoring in money laundering respect and terrorism financing risks, processes and procedures of risk analysis internal complex should be organically integrated into the bank's credit policy; 2) bank risks are minimized through risk management organization in such a way as to adequately identify and evaluate risks by bank personnel, as well as to evaluate risks of client's creditworthiness and type of credit operation; 3)

availability of personnel-oriented credit risks management system will allow identifying existing risks in accordance with the requirements of external and internal risk monitoring, defining the danger's level and, applying certain methods and techniques of risk management, reducing their negative impact on bank activity; 4) transformation of the approach to risk assessment and creation of a risk prevention system from financially oriented to personnel-oriented, which emphasizes the defining the person role in the bank credit risks management system.

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